

High alpha-linolenic acid flaxseed (Linum usitatissimum): some nutritional properties in humans.

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Although high alpha-linolenic acid flaxseed (*Linum usitatissimum*) is one of the richest dietary sources of alpha-linolenic acid and is also a good source of soluble fibre mucilage, it is relatively unstudied in human nutrition. Healthy female volunteers consumed 50 g ground, raw flaxseed/d for 4 weeks which provided 12-13% of energy intake (24-25 g/100 g total fat). Flaxseed raised alpha-linolenic acid and long-chain n-3 fatty acids in both plasma and erythrocyte lipids, as well as raising urinary thiocyanate excretion 2.2-fold. Flaxseed also lowered serum total cholesterol by 9% and low-density-lipoprotein-cholesterol by 18%. Changes in plasma alpha-linolenic acid were equivalent when 12 g alpha-linolenic acid/d was provided as raw flaxseed flour (50 g/d) or flaxseed oil (20 g/d) suggesting high bioavailability of alpha-linolenic acid from ground flaxseed. Test meals containing 50 g carbohydrate from flaxseed or 25 g flaxseed mucilage each significantly decreased postprandial blood glucose responses by 27%. Malondialdehyde levels in muffins containing 15 g flaxseed oil or flour/kg were similar to those in wheat-flour muffins. Cyanogenic glycosides (linamarin, linustatin, neolinustatin) were highest in extracted flaxseed mucilage but were not detected in baked muffins containing 150 g flaxseed/kg. We conclude that up to 50 g high-alpha-linolenic acid flaxseed/d is palatable, safe and may be nutritionally beneficial in humans by raising n-3 fatty acids in plasma and erythrocytes and by decreasing postprandial glucose responses.